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MILITARY SPECIFICATION

GLOVES, CHEMICAL AND OIL PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for unsupported, dipped rubber gloves, with a cotton-flock lining.

1.2 Classification. The gloves shall be procured in the following sizes as specified (see 6.2):

Schedule of Sizes
7, 8, 9, 10, 11

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard
NN-P-71	- Pallet, Material Handling, Wood, Stringer, Construction, 2 Way and 4 Way (Partial)

MILITARY

MIL-O-82672	- Otto Fuel II
MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry

STANDARDS

FEDERAL

FED-STD-601	- Rubber, Sampling and Testing

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-168	- Visual Inspection Guide for All Rubber Gloves Except Surgical

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

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(Applications for copies may be addressed to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D-412 Test for Tensile Strength

D-471 Test for Change in Properties
of Elastomeric Vulcanizates Resulting
from Immersion in Liquids

D-573 Test for Air Heat Test, Air Heating Medium

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

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3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Rubber Compounds. The gloves shall be made of a carboxylated butadiene-acrylonitrile rubber latex, suitably compounded without the use of plasticizers or inert fillers, and vulcanized to meet the requirements specified herein (see 4.4.1).

3.3.1.1 Color. Unless otherwise specified by the procuring activity, the approved color of the finished gloves shall be either white or in light shades of green or blue (see 6.5).

3.3.2 Lining. Each finished glove shall have a cotton flock lining suitably adhered to the rubber compound on the inside (see 4.4.1 and 6.4).

* 3.4 Label. Each finished glove shall have a combination identification and cautionary statement inscription as shown below, legibly marked and centered on wrist of palm side, with the bottom of the label 3/4 to 7/8 inch from the bottom edge of cuff. The inscription shall be made either by printing, stenciling, or stamping. The size of the lettering for the identification information shall be not less than 1/8 inch in height. The cautionary statement shall be printed in larger letters than all other lettering in the label. The marking ink shall be of any permanent contrasting color, provided it is consistent with the approved color requirements of the finished glove (see 3.3.1.1):

CONTRACT NO: DLA 100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE: 9 (EXAMPLE)

FOR PROTECTION AGAINST USN OTTO FUELS, DO NOT REUSE

3.5 Physical and chemical requirements. The finished gloves shall conform to the physical requirements shown in Table I when tested as specified in 4.5.

Table I - Physical and Chemical Requirements of Finished Gloves

Characteristic	Requirement	
	Minimum	Maximum
Thickness, inch: Cuff, palm, two crotch and two fingertip areas	.013	-
Tensile strength, p.s.i.		
Original	3000	-
After aging	2700	-
Ultimate elongation, percent		
Original	380	-
After Aging	360	-
Area change, percent after immersion in Fuel B	-	35.0
Porosity	Pass 1/	-

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Characteristic	Requirement	
	Minimum	Maximum
Resistance to permeation by:		
Otto fuel: Permeation rate at 2 hrs, micrograms/minute	-	.10
Cumulative permeation, 2 hrs period, micrograms	-	1.0

1/ There shall be no evidence of air bubbles.

3.6 Dipping forms. The dipping forms utilized by the contractor in his manufacturing process may be made from any commercially designed mold, provided that the gripping surface of the finished gloves (palm and fingers) has a non-slip embossed pattern design.

* 3.7 Design. The gloves shall be of a standard four finger and thumb style, with either a pinked, natural dipped or rolled bottom cuff edge (see Figure 1).

3.7.1 Figures. Figures 1 through 5 are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.8 Chlorination. The rubber surfaces of each finished glove shall be chlorinated to reduce tackiness. Excessive chlorination shall be avoided to prevent excessive stiffening, discoloration, or slipperiness (see 4.4.1).

* 3.9 Measurements. The sizes and measurements of the finished gloves shall conform to the dimensions as specified in Table II. All measurements and tolerances are expressed in inches and shall be taken with the glove flattened without pulling or stretching.

Table II - Finished Measurements

Measurement	Size					Tolerance
	7	8	9	10	11	
Overall Length (A)	13	13	13	13	13	Minimum
Width of palm (B)	3 7/8	4	4 3/8	4 7/8	5 3/8	± 1/4
Width of wrist (C)	3 1/4	3 5/8	3 7/8	4 3/4	5 1/8	Minimum
Width of cuff (D)	4 1/2	4 3/4	5 1/4	5 5/8	6	± 1/4

NOTE: A through D refer to Figure 1.

A. Measurement shall be taken on back side from tip of second finger to top open edge of glove.
B. Measurement shall be taken across the back at thumb crotch, folded edge to folded edge.
C. Measurement shall be taken across the narrowest part of the wrist on back side, folded edge to folded edge.
D. Measurement shall be taken across the top open edge of glove on back side, folded edge to folded edge.

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3.10 Workmanship. The finished gloves shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 4. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2.1 and 4.4.2.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be submitted for the material identification of the rubber compound (see 3.3.1), the cotton flock lining for the inside of the gloves (see 3.3.2), and for the required chlorination process of the finished gloves (see 3.8).

4.4.2 Examination of the end item. Examination of the end item shall be in

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accordance with 4.4.2.1 through 4.4.2.3. The sample unit shall be one completely fabricated glove, and selection shall be made in pairs. Pairing defects shall be scored as a single defect. The lot size shall be expressed in units of one glove each.

* 4.4.2.1 Visual examination.

Defect	Classification Major	Minor
Pairing		
Gloves not mated in pairs e.g., two left, two right gloves submitted as a pair, not same size	X	
Definite variation in color or appearance		X
Design		
Any characteristic not in accordance with specified requirements (unless otherwise classified herein)	X	
Color		
Not as specified		X
Cleanness		
a. Grease, oil stain, or dirt on glove which cannot be easily removed	X	
b. Not clean, i.e., dirty, but dirt can be easily removed		X
Construction and Workmanship (applicable to inside and outside of glove)		
a. Any cut, hole, tear, rip or rupture through material	X	
b. Evidence of improper vulcanization, e.g., tacky, etc.	X	
c. Any repair or patch	X	
d. Any blister, open or closed	X	
e. Any pinch, thin spot, pit, abraded area, deep crease, or similar defective condition resulting in a thickness at the defect less than the minimum thickness allowed	X	
f. Any burned spot	X	
g. Any malformation or distortion:		
- Seriously affecting serviceability	X	
- Affecting serviceability, but not seriously		X
h. Any solid rubber ridge, run, or lump		X
i. Not constructed as specified, i.e., bottom open edge of glove not rolled, natural edge or pinked as specified	X	
j. Bottom rolled edge (when applicable) of glove crooked or loose		X
k. Length not as specified	X	
l. Cotton flock lining omitted on inside of glove	X	

NOTE: For rubber defect illustrations, see MIL-STD-168 Markings

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Defect	Classification Major	Minor
Marking		
a. Missing, incorrect or illegible	X	
b. Misplaced, height of characters not as specified, or not accomplished as specified		X
c. Not permanent, i.e., can be easily rubbed off with a moistened thumb		X

4.4.2.2 Dimensional examination. The gloves shall be examined for dimensional defects as specified in 3.9. Any dimension that is not within the established tolerance shall be classified as a defect.

4.4.2.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection level	AQL
For defects applicable to 4.4.2.1	II	2.5
For defects applicable to 4.4.2.2	S-3	4.0

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container, open and noncontinuous heat sealed seams of polyethylene bag, or incorrectly fabricated bag.
Content	Number of paired gloves per shipping container is more or less than specified. 1/ Size shown on one or more items not as specified on shipping container. 2/

1/ For this defect, two (2) shipping containers in the sample shall be examined.

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2/ For this defect, one item from each shipping container in the sample shall be examined.

* 4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

4.5 Testing of the end item. Each lot of the end item shall be tested for the characteristics specified in Table III and in accordance with FED-STD-601 and ASTM test methods, whenever applicable. The lot size shall be expressed in units of one glove each, and selection shall be made in pairs. All requirements shall be applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in Table III, they shall be as required by the referenced test method. All test reports shall contain the individual values used in expressing the final result.

a. Government acceptance test (see 6.6). The government shall perform an acceptance test for resistance to permeation by Otto fuel. The sample unit for this test shall be one pair, and the inspection level shall be S-2. This test characteristic is considered critical and any failure shall be cause for rejection of the lot represented by the sample.

b. Porosity test. The sample unit for this test shall be one pair and the inspection level shall be S-2. This test characteristic is considered critical and any failure shall be cause for rejection of the lot represented by the sample.

c. All tests other than Government acceptance and porosity tests. For all remaining tests, the sample unit shall be six (6) pairs of gloves. The sample size shall be as specified below. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified.

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Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

d. The Government reserves the right to draw test samples for each lot or perform 100% testing to verify the contractors test results, and to defer acceptance pending completion of Government verification testing.

Table III - End item testing

Characteristic	Requirement Paragraph	Test Method	No. of determinations per sample unit	Results Reported As
Thickness: Cuff, palm, two crotch and two finger areas	3.5	2011 1/	12 in each of the designated areas	Average measurement in each area to the nearest 0.001 inch
Tensile strength: Original	3.5	ASTM D412 2/	-	-
After aging	3.5	ASTM D573 2/	-	-
Ultimate elongation: Original	3.5	ASTM D412 2/	-	-
After aging	3.5	ASTM D573 2/	-	-
Area change: After immersion in Fuel B	3.5	ASTM D471 3/	-	-
Porosity	3.5	4.5.1	2	"Pass" or "Fail"
Resistance to permeation by Otto fuel	3.5	4.5.2	3	"Pass" or "Fail"

1/ The combined thickness (rubber and flocking) shall be determined at each of these areas on the smooth portions.
2/ The test specimens shall be dumbbell shaped conforming to Die C. The combined thickness (rubber and flocking) shall be used in calculating tensile strength values. The test specimens shall be cut so that the lengthwise portion of the specimen will be in the mill (dipping) direction and vertically in line with the fingers.
3/ The area change method shall be used and Fuel B (mixture of isooctane 70% and toluene 30%) as the immersion medium specified in paragraph 11 and Table

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and Table II respectively. Measure length and width of sample before and after immersion in Fuel B for 24 hours. Calculate percent area change as follows:

$$\% \text{ Area change} = \frac{(L)(V) - (Lo)(Vo) \times 100}{(Lo)(Vo)}$$

Lo = Initial length
Vo = Initial width
L = Length after immersion
V = Width after immersion

4.5.1 Porosity.

4.5.1.1 Test apparatus. The test apparatus shall be of a mechanism as described below or any other similar type in principle, provided the results can be obtained as specified in 4.5.1.2.

A hollow wooden shank, 3 inches in length, tapered on the outer surface to fit the open end of the rubber glove and a steel ring tapered on the inner surface to fit over the wooden shank shall be used. The shank shall be attached to a steel base plate to make an airtight joint, and the tapered ring shall be bolted to the base plate so that it can be tightened over the shank. The base plate shall be fitted with 1/4 inch internal pipe size brass nipple which shall be connected with other pipe fittings to mount a tire inflation valve, an air pressure gauge, and a pressure release valve (see Figure 2).

4.5.1.2 Procedure. The open end of the rubber glove shall be pulled over the shank of the test apparatus and clamped firmly in place by tightening the steel ring against the base plate. Compressed air shall be fed into the glove to a pressure of 0.50 (± .25) p.s.i. The pressurized glove shall be immersed in water for 1 minute for observation of porosity as indicated by air bubbles. No visible bubbles shall be permitted within the 1 minute period.

4.5.2 Resistance to permeation by Otto fuel.

4.5.2.1 Scope. This test method evaluates the resistance of the rubber glove materials to permeation by the propylene glycol dinitrate (PGDN), a component of Otto fuel, utilizing an electron capture gas chromatograph equipped with a heated gas sample valve. An alternate procedure may be used by the testing agency, provided that comparable results to those obtained by the method described below can be produced. The fuel used in this test should be Otto fuel conforming to MIL-O-82672.

4.5.2.2 Definitions. Otto fuel is a liquid monopropellant, composed of nitrate ester (1, 2 propylene glycol dinitrate) to which a desensitizer and stabilizer have been added.

* 4.5.2.3 Precautionary note. Because of the toxicity of Otto fuel, precautions should be taken against its inhalation and contact with the skin as described in NAVSEA S-6340-AA-MMA-010. Reference should be made to the Commander, Naval Medical Command (MEDCOM 422), Washington, D.C. 20372-5120 for the latest threshold limit value established for exposure to Otto fuel.

4.5.2.4 Apparatus (See Figures 3 and 4). The test apparatus shall consist

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of the following:

(a) Sample holder. The sample holder, consisting of two (2) flanged cups with inside diameter 1.5 to 2.5 cm between which the test specimen can be clamped, shall be made of glass or Teflon (FEP or TFE) or any other material that does not absorb PGDN vapor. The upper flanged cup shall have a cover to prevent evaporation of the test liquid. The lower chamber shall be provided with an inlet on the bottom to direct air across the bottom surface of the test specimen, to transport the PGDN permeating the test specimen for testing (see Figure 3).

(b) Electron Capture Gas Chromatograph. This device shall be equipped with a heated gas sample valve, which shall be controlled at 100 ± 10°C. The sample valve shall be capable of being manually or automatically actuated to sample airflow every 5 minutes.

(c) Recording device. A strip chart recorder to indicate the gas chromatograph signal.

(d) Vacuum pump. To draw a flow of air from the lower chamber of the sample holder to the sample valve.

(e) Flowmeter. To measure the airflow rate through the sample holder.

(f) Charcoal filter. To prevent PGDN from contaminating test area.

4.5.2.5 Test specimens. Three test specimens shall be cut from each glove being tested; one each from the thumb, index finger, and palm. Each specimen shall be sufficiently large to cover the effective area of the sample holder, and shall be cut so that the embossed or roughened portion of the glove is in the center of the sample facing upward.

4.5.2.6 Number of determinations. One determination shall be made on each test specimen.

4.5.2.7 Procedure.

(a) Calibrate gas chromatograph with known vapor concentration of PGDN vapor in air generated by a diffusion-tube-type calibration system (see Figure 5).

(b) Place test specimen in sample holder with roughened or embossed surface facing outward. Provisions shall be made to control the temperature of sample holder to 22 ± 2°C. The sample holder must be kept in fume hood during test.

(c) Connect sample valve to sample holder as shown in Figure 4.

(d) Measure and record airflow rate through sample holder.

(e) Place 2.0 microliters (2.5 milligrams) of Otto fuel on the upper surface of the test specimen. The liquid should be spread over a circular area no greater than 0.8 cm in diameter in the center of

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the test specimen. Place cover on sample holder. Air passing through the lower chamber is sampled for PGDN permeating through the test specimen.

(f) Actuate sample valve at 5-minute intervals for 2 hours to measure and record concentrations of PGDN in micrograms per liter.

4.5.2.8 Calculations.

(a) Permeation rate. Calculate the permeation rate in micrograms per minute at 5-minute intervals up to 2 hours as follows:

Permeation rate = PGDN concentration x flow rate where permeation rate is in micrograms per minute. PGDN concentration is in micrograms per liter and flow rate is in liters per minute.

(b) Cumulative permeation. Calculate the permeation in micrograms during each 5-minute period as follows:

$$P_n = 2.5 (R_n + R_{n+1})$$

where P_n = the permeation during the nth period in micrograms

R_n = the permeation rate at the beginning of the nth period in micrograms per minute

R_{n+1} = the permeation rate at the end of the nth period

Calculate the cumulative permeation as the sum of P_n's from the start of the test up to the nth period for each 5-minute period up to 2 hours.

4.5.2.9 Report.

(a) Permeation rate. Report permeation rate at two hours to nearest 0.01 micrograms per minute. Report number of 5-minute intervals that the rate exceeds 0.10 micrograms per minute if less than total.

(b) Cumulative permeation. Report cumulative permeation at two hours to nearest 0.1 micrograms. Report elapsed time when cumulative permeation exceeds 1.0 micrograms if less than 2 hours.

4.5.2.10 Notes.

(a) Advisory information. The following test equipment has been used successfully for the permeation resistance test method:

1. Gas chromatograph. Analytical Instruments Development Model 510 or 530 with heated Carle Cat. No. 5518 sample valve, 0.2 ml FEP Teflon sample loop, and Carle Model 2050 valve actuator.

2. Calibration system. Calibration System Model 390, Analytical Instrument Development, Inc., Route 41 and Newark Road,

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Avondale, PA 19311. Fittings and tubing that come into contact with PGDN vapors in the calibration system were replaced with fittings and tubing of polytetrafluoroethylene (Teflon, or equal). A diffusion tube in accordance with Figure 5 was used with calibration system.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A. Each pair of gloves shall be inserted in a flat style, clear polyethylene film bag, of 0.0015 inch thickness (± 20%) tolerance. The polyethylene bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag, and may be fabricated from polyethylene film tubing or sheeting. The final closure of the bag shall be heat sealed, with the seal made as close as possible to the open end. Prior to or during the closure operation, excess air within the bag shall be expelled.

5.1.2 Level C. Gloves shall be preserved-packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Forty-five (45) pairs of gloves of one size only and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF, class weather-resistant, variety DW, grade V15c, style RSC, of PPP-B-636. The fiberboard for the box liner shall conform to type CF, class domestic of PPP-B-636. Level A packages shall be packed flat, three in length, one in width, and fifteen in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Containers shall be closed in accordance with Method III and reinforced in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B. Forty-five (45) pairs of gloves of one size only and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF, class domestic, variety DW, grade 200, of PPP-B-636. The fiberboard for the box liner shall conform to type CF, class domestic of PPP-B-636. Level A packages shall be packed flat, three in length, one in width, and fifteen in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Containers shall be closed in accordance with Method III and reinforced in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Gloves, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail

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distribution. Containers shall comply with US Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

* 5.2.4 Palletization. When specified (see 6.2) item, packed as specified shall be palletized on a 4-way entry pallet in accordance with load type IA of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1, fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.3 Marking In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

* 5.3.1 Polyethylene bagged packaged Polyethylene bagged packages shall have the following information listed below, legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag, so as to permit ready identification:

GLOVES, CHEMICAL AND OIL PROTECTIVE
CONTRACT NO:
STOCK NO:
SIZE:
QUANTITY:

6. NOTES

6.1 Intended use. The gloves covered by this specification are intended to be used for the protection of personnel against exposure to toxic Otto fuel and other chemicals involved in the fueling, cleaning, maintenance, and damage control of the Navy Mark 48/46 torpedoes.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes required (see 1.2).
c. When first article inspection is required, (see 3.2) the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, quantity, and testing and approval.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.2.4).

6.3 Samples. For access to samples address the procuring activity issuing the invitation for bids.

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6.4 Flocking. The utilization of a "Flock Booth" whereby the cotton flock is mechanically blown onto the rubber compound for the inside glove lining has been found to be a suitable procedure (see 3.3.2).

6.5 Approved colors. Approved colors are those which do not mask the contamination of the gloves by the reddish orange Otto fuel, the torpedo propellent. Examples of non-acceptable colors are black, orange, red and yellow.

* 6.6 Government acceptance test (see 4.5a). Government acceptance tests shall be performed at the Defense Personnel Support Center, 2800 So. 20th St., Attn: DPSC-FSL, Philadelphia, PA. 19101-9965.

* 6.7 Die C. Dies conforming to ASTM D412 and D573 may be obtained from the following suggested sources:

Boston Cutting Die Company
50-52 Freeport Street
Boston,Ma 02122
Tel: (617) 288-9400

Philadelphia Consolidated Die Company
327 Race Street
Philadelphia, Pa 19103
Tel: (215) 627-2109

In case of dispute, dumbell specimens shall be cut from paper or oil board and measured to determine conformance to the Die C measurements listed in ANSI/ASTM TM 412 and TM 573.

* 6.8 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval.

* 6.9 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.10 Subject term (key word) listing.
Chemical and Oil Protective
Color, green, blue or white
Latex, rubber
Style, four finger and thumb
Utility

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Custodian:
Navy - NU
Army - GL

Review Activity:
DLA - CT
Navy - OS
Army - MD, GL

Preparing Activity:
Navy - NU

Project No. 8415-0536

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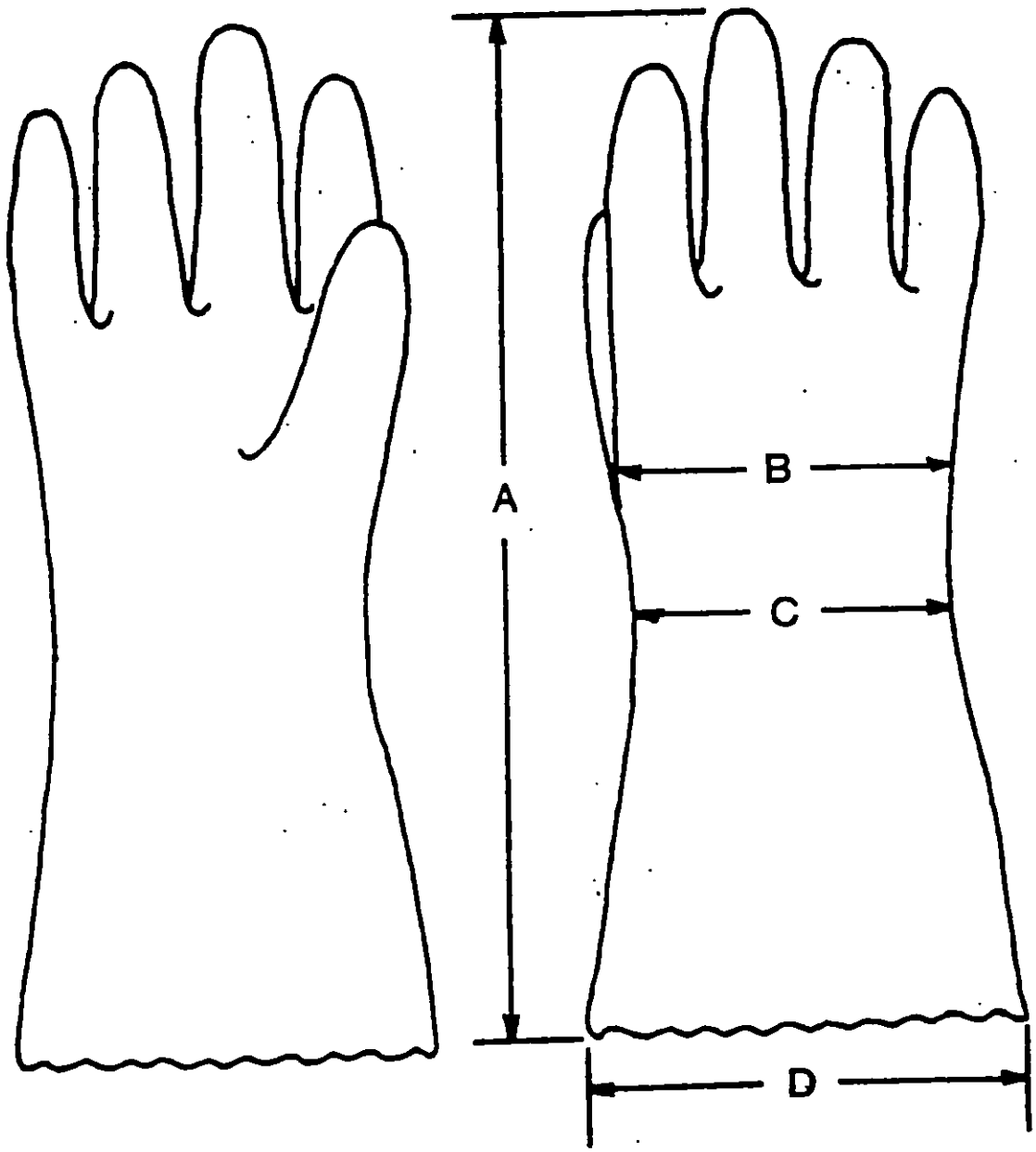
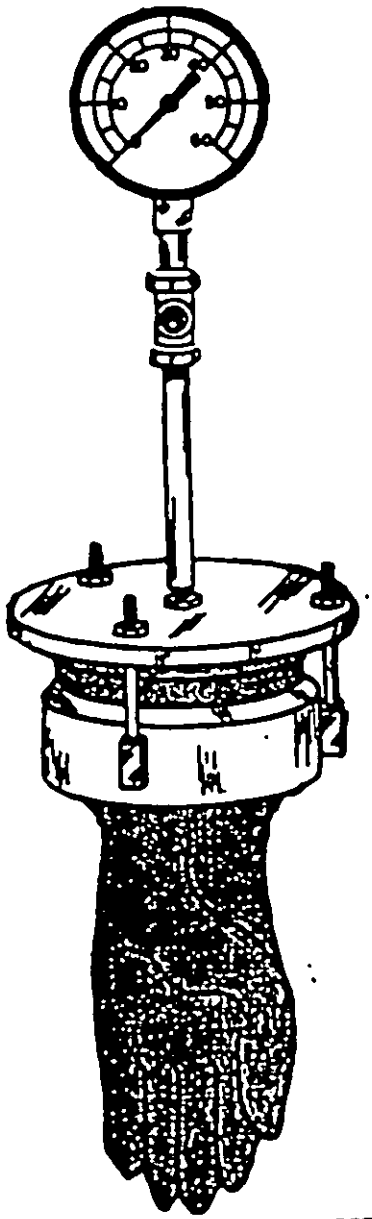


Figure 1 - Gloves, Chemical and Oil Protective

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APPARATUS FOR POROSITY TEST

Figure 2 - Gloves, Chemical and Oil Protective

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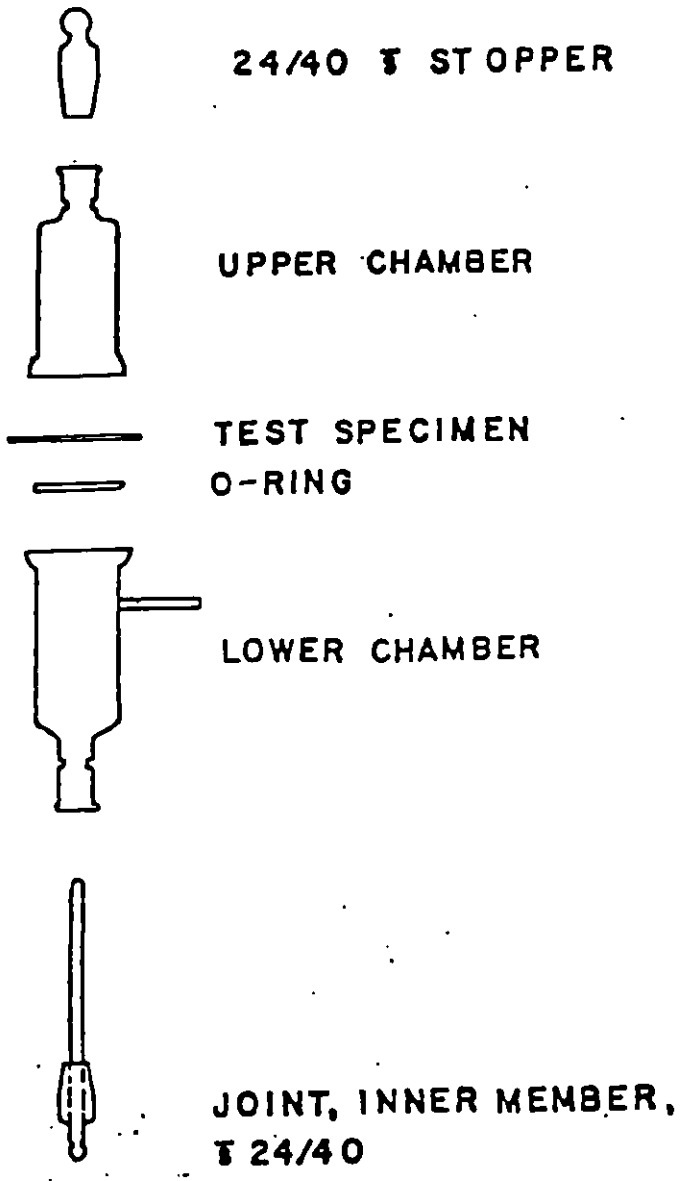
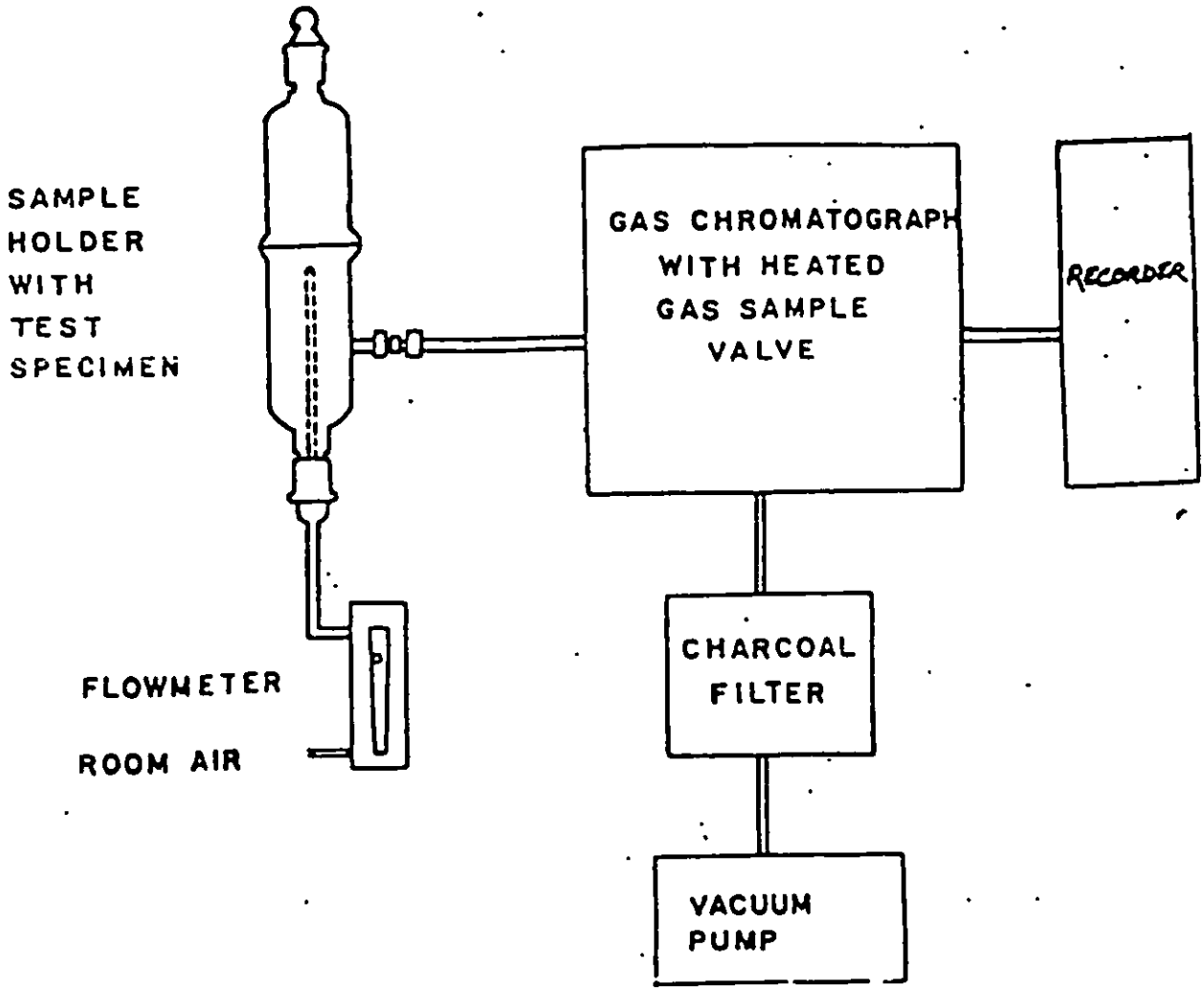


Figure 3 - Gloves, Chemical and Oil Protective

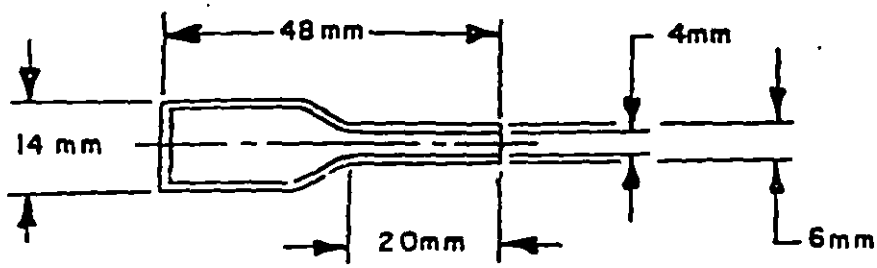
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TEST APPARATUS

Figure 4 - Gloves, Chemical and Oil Protective

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MATERIAL : GLASS
ALL DIMENSIONS IN MILLIMETERS

TEMPERATURE	DIFFUSION RATE	FLOW RATE	PGDN CONCENTRATION	
°C	µg/min	l/min	µg/l	ppm
50	1.30	1.88	0.69	0.10
50	1.30	0.94	1.38	0.20

OTTO FUEL DIFFUSION TUBE

Figure 5 - Gloves, Chemical and Oil Protective

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